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Circularly Polarized Scattering Radiation from Silicon Nanosphere

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Abstract

A dielectric nanosphere with orthogonal electric dipole (ED) and magnetic dipole (MD) Mie resonances can be a nanoantenna radiating circularly polarized light in specific directions if the amplitudes and the phase relations are properly designed. We first show from theoretical calculations that a silicon nanosphere illuminated with linearly polarized plane wave radiates circularly polarized light at the wavelength in between the ED and MD resonances if the refractive index of a surrounding medium (n_m) is around 1.3; the ellipticity of the scattered light can be >0.99 when n_m is in a 1.19-1.35 range. We then prepare size-purified silicon nanospheres suspended in water (n_m =1.33) and study the angle- and circular-polarization-resolved scattering spectra. We experimentally demonstrate that circularly polarized light is radiated in specific directions under linearly polarized plane wave illumination. We also show that the wavelength of the radiation of circularly polarized light can be controlled in the whole visible range by controlling the silicon nanosphere diameter in 100-200 nm range.

1. Introduction

A nanoparticle with multiple electromagnetic eigenmodes can be a subwavelength-size antenna that radiates circularly polarized light if the orientations, the amplitudes and the phase relations of the modes are properly designed[1–8]. For example, in a plasmonic split ring resonator with coupled electric dipole (ED) and electric quadrupole (EQ) modes, emission of circularly polarized light in specific directions from dipole emitters placed on top has been demonstrated[5]. The observed maximum ellipticity of the emission reaches ±~0.5. As a nanoantenna for circularly polarized radiation, a nanoparticle of high-refractive index dielectrics such silicon have several advantages compared to plasmonic ones. Since dielectric nanoparticles inherently have multiple electric and magnetic Mie modes, they have a greater degree of freedom to design the antenna property. The interference between these modes brings about rich phenomena such as well-known Kerker effect[9,10]. If the ED and the magnetic dipole (MD) modes are in phase and orthogonal in space and have the equal oscillation amplitude, scattering in backward direction is perfectly suppressed (1st Kerker condition)[11–13]. At the 1st Kerker condition, the helicity of incident light is preserved, i.e., if circularly polarized light is incident to a silicon nanosphere, the circular polarization is fully preserved not only in the far field but also in the near field[14–23][24]. On the other hand, if the ED and MD modes are π out-of-phase, scattering in the forward direction is suppressed (2nd Kerker condition)[13,21,23], and the helicity of circularly polarized incident light is flipped[14,17,20,24].

More complicated control of scattering radiation can be made by excitation of specific Mie modes in a specific direction by using artificially designed structured beam[25]. For example, excitation of an ED mode parallel to the wavevector of incident light by a radially polarized structured beam results in unidirectional light scattering in the transverse direction[26,27]. Excitation of ED and MD modes parallel in space and π/2 phase difference generates a chiral dipole that emits circularly polarized light in all directions[28]. Selective excitation of specific Mie modes and emission of circularly polarized light along specific directions is also made by electron beam irradiation of a silicon nanoparticle[29].

In a silicon nanoparticle, orthogonal ED and MD modes can be excited simply by irradiating linearly polarized light as schematically shown in **Figure 1**. If the phase difference of the modes is π/2, circularly polarized light is emitted in the direction where the scattering intensity of the two modes is the same. Therefore, an achiral nanoantenna that radiates circularly polarized light into specific directions can be realized without using structured beam and without introducing perturbation to the shape. In this paper, we discuss the possibility that a silicon nanosphere can radiate circularly polarized light under linearly polarized plane wave illumination. We first show that the concept in Figure 1 does not work in a silicon nanosphere in air due to the deviation of the phase difference from π/2 at the wavelength of equal ED and MD scattering intensity. As a method to control the phase difference, we focus on the refractive index of a surrounding medium (n_m). We show that the phase difference can be effectively controlled in a wide range by n_m and the ellipticity of the scattered light can be >0.99 when n_m is in a 1.19-1.35 range. Finally, we experimentally prove the concept by measuring angle dependence of the degree of circular polarization of scattered light ($\eta(\varphi)$) for water solutions of size-purified silicon nanospheres. Experimentally observed η reaches ~0.8.

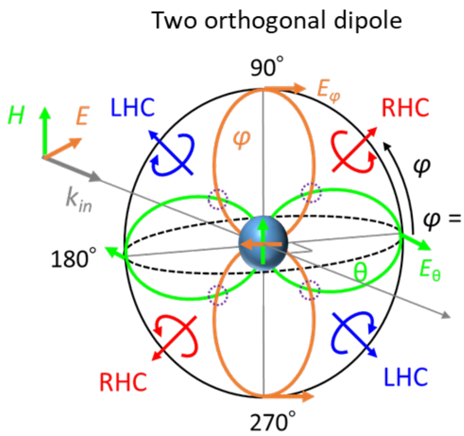

Figure 1. Scattering radiation pattern of a dielectric nanoparticle when orthogonal ED and MD modes are excited by linearly polarized plane wave illumination. When the amplitude of the two modes is equal and the phase difference is ±π/2, circularly (RHC or LHC) polarized light is emitted in specific directions (φ = π/4, 3π/4, 5π/4, 7π/4 in the transverse direction).

2. Analytical model

2.1. Light Scattering by a Silicon Nanosphere in Air

We first analytically calculate the polarization property of a silicon nanosphere under linearly polarized plane wave illumination. **Figure 2**(a) shows the definition of the coordinate system. The incident light polarized in the *x*-direction propagates in the +*z* direction. From the Mie theory, the electric field of the scattered light along the φ and θ directions at a position (r, θ, φ) is

$$E_\varphi(\theta,\varphi) = \frac{\sin\varphi}{\rho}\sum_{n=1}^{\infty} E_n\big(b_n\xi_n\tau_n(\cos\theta) - i a_n\xi_n'\pi_n(\cos\theta)\big) \tag{1}$$

$$E_\theta(\theta,\varphi) = \frac{\cos\varphi}{\rho}\sum_{n=1}^{\infty} E_n\big(i a_n\xi_n'\tau_n(\cos\theta) - b_n\xi_n\pi_n(\cos\theta)\big),$$

respectively, where a_n and b_n are the Mie coefficients, $\rho = kr$, k is the free space wavenumber, $E_n = i^n E_0(2\mathrm{n}+1)/\mathrm{n}(\mathrm{n}+1)$, ξ_n is the Riccati−Bessel function, $\pi_n = P_n^1/\sin\theta$, and $\tau_n = dP_n^1/d\theta$[30]. If we consider only the ED (a_1) and MD (b_1) modes and scattering in the $\theta = \pi/2$ direction, Eq. (1) is simplified to

$$E_\varphi\left(\frac{\pi}{2},\varphi\right) = \frac{\sin\varphi}{\rho}E_1(-i a_1\xi_1') \tag{2}$$

$$E_\theta\left(\frac{\pi}{2},\varphi\right) = \frac{\cos\varphi}{\rho}E_1(-b_1\xi_1).$$

Since $E_\varphi(\pi/2,\varphi)$ contains only the a_1 Mie coefficient, it corresponds to the ED mode. On the other hand, $E_\theta(\pi/2,\varphi)$ contains only the b_1 Mie coefficient and corresponds to the MD mode (see Figure S1 in Supporting Information for more details).

By using $E_\varphi(\pi/2,\varphi)$ and $E_\theta(\pi/2,\varphi)$, the degree of circular polarization of light scattered in the $\theta = \pi/2$ direction is expressed as

$$\eta\left(\frac{\pi}{2},\varphi\right) = \frac{2\left|E_\varphi\left(\frac{\pi}{2},\varphi\right)\right|\left|E_\theta\left(\frac{\pi}{2},\varphi\right)\right|\cos\left(\varphi_{MD} - \varphi_{ED} \pm \frac{\pi}{2}\right)}{E_\varphi\left(\frac{\pi}{2},\varphi\right)^2 + E_\theta\left(\frac{\pi}{2},\varphi\right)^2}, \tag{3}$$

where $\varphi_{MD} = \arg(b_1)$ and $\varphi_{ED} = \arg(a_1)$ are the phases of the ED and MD modes, respectively. If the amplitude of the ED and MD modes is the same ($|a_1| = |b_1|$), $|E_\varphi(\pi/2,\varphi)| = |E_\theta(\pi/2,\varphi)|$ at φ = π/4, 3π/4, 5π/4, 7π/4. Therefore, if the phase difference between the ED and MD modes ($\varphi_{MD} - \varphi_{ED}$) is $\pm\pi/2$, $\eta(\theta = \pi/2, \varphi = \pi/4, 3\pi/4, 5\pi/4, 7\pi/4)$ becomes ±1, and thus circularly polarized light is emitted under linearly polarized light illumination (Figure S1 in Supporting Information).

We examine whether a silicon nanosphere in air can satisfy above mentioned conditions or not. Figure 2(b) shows the scattering cross section spectrum of a silicon nanosphere 150 nm in diameter in air. We consider only the ED and MD modes in the calculation and ignore the higher-order modes. The contributions of the ED and MD modes to the spectrum are also shown in Figure 2(b). Figure 2(c) shows the phases of the modes and the difference ($\varphi_{MD} - \varphi_{ED}$). The wavelength where the intensities of the ED and MD modes are the same and the phase difference is zero at the long wavelength side of the two resonances corresponds to the 1st Kerker ($1st^1$) condition. On the other hand, the wavelength where the intensities are the same in between the two resonances is the 2nd Kerker ($2nd^1$) condition, although the phase difference deviates from the ideal value of π. Also, at the short wavelength side of the ED and MD resonances, the intensities of the ED and MD modes become equal, and we label them $1st^2$ and $2nd^2$. As can be seen in Figure 2(c), the phase difference of π/2 is not satisfied at all the four wavelengths where the scattering intensities of the ED and MD modes are the same. Therefore, emission of circularly polarized light schematically shown in Figure 1 does not occur in a silicon nanosphere in air.

2.2. Light Scattering by a Silicon Nanosphere in Water

In order to control the phase difference, we immerse a silicon nanosphere in a medium with different refractive indices (n_m). Figure 2(d) shows the phase difference at the wavelengths where the ED and MD scattering intensities are the same as a function of n_m. The phase difference depends strongly on n_m except for the 1st Kerker condition ($1st^1$). Especially, at the $2nd^1$ wavelength, the phase difference crosses π/2 around n_m =1.3. Figure 2(e) shows the degree of circular polarization ($\eta(\theta = \pi/2, \varphi = \pi/4)$) at $2nd^1$ as a function of n_m. $\eta(\theta = \pi/2, \varphi =$

π/4) is larger than 0.99 for $1.19 < n_m < 1.35$. Therefore, if we immerse a silicon nanosphere in a medium having the refractive index in that range, it works as a nanoantenna that radiates circularly polarized light under linearly polarized plane wave illumination. A typical material in the refractive index range is water (n_m=1.33).

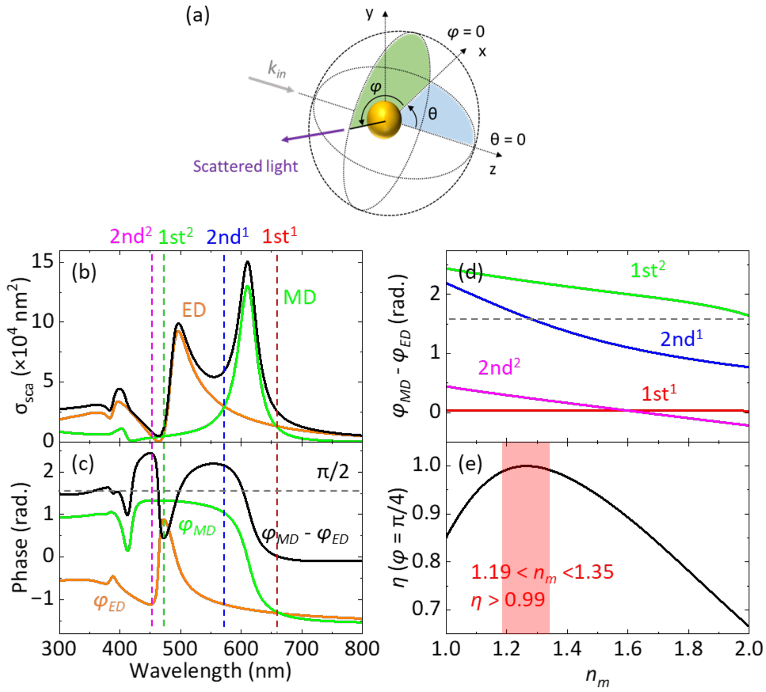

Figure 2. a) Definition of the angles of the incident and scattering light. The incident light propagates in the positive direction of the z-axis, and the direction of the scattered light is defined in the spherical coordinate system, θ ($0 \leq \theta \leq \pi$) and φ ($0 \leq \varphi \leq 2\pi$). b) Scattering cross section spectrum of a silicon nanosphere 150 nm in diameter. Contributions of ED and MD modes are also shown. c) Phases of ED and MD modes and the phase difference as a function of wavelength. The refractive index of the medium is $n_m = 1.00$. d) Phase difference of ED and MD modes at the wavelengths of the same scattering intensity as a function of n_m. The horizontal dashed line indicates the desired phase difference (π/2). e) Degree of circular polarization of scattered light in the $\theta = \pi/2$ and $\varphi = \pi/4$ direction ($\eta(\theta = \pi/2, \varphi = \pi/4)$) as

a function of n_m at the wavelength of 2nd^1. In the red region ($1.19 < n_m < 1.35$), $\eta(\theta = \pi/2, \varphi = \pi/4)$ exceeds 0.99.

Figure 3(a) and (b) shows the scattering cross section spectrum and the phase spectrum, respectively, of a silicon nanosphere 150 nm in diameter in water, and Figure 3(c) shows the degree of circular polarization in the $\varphi = \pi/4$ and $3\pi/4$ directions ($\theta = \pi/2$). As expected, at 2nd^1, $\eta(\theta = \pi/2, \varphi = \pi/4, 3\pi/4)$ becomes ± 1. Figure 3(d) shows the scattering radiation pattern at the 2nd^1 wavelength. RHC polarized light is emitted in the $\varphi=\pi/4$, $5\pi/4$ directions, while LHC polarized light in the $\varphi=3\pi/4$, $7\pi/4$ directions. Figure 3(e) shows the degree of circular polarization as a function of wavelength and φ. Circularly polarized light is emitted in the 500-600 nm range. In Figure 3(c) and 3(e), we notice that circularly polarized light is also emitted below 450 nm. However, this does not occur in actual silicon nanosphere samples shown later because of the contributions of the higher-order modes.

Figure 3(f) shows the wavelength of the 2nd Kerker condition (2nd^1) as a function of silicon nanosphere diameter. It shifts to longer wavelength with increasing the diameter. In the 100-200 nm diameter range, i.e., in the wavelength range of 450-700 nm, $\eta \cong 1$ can be achieved. More detailed information on the size dependence is shown in Supporting Information (Figure S2)

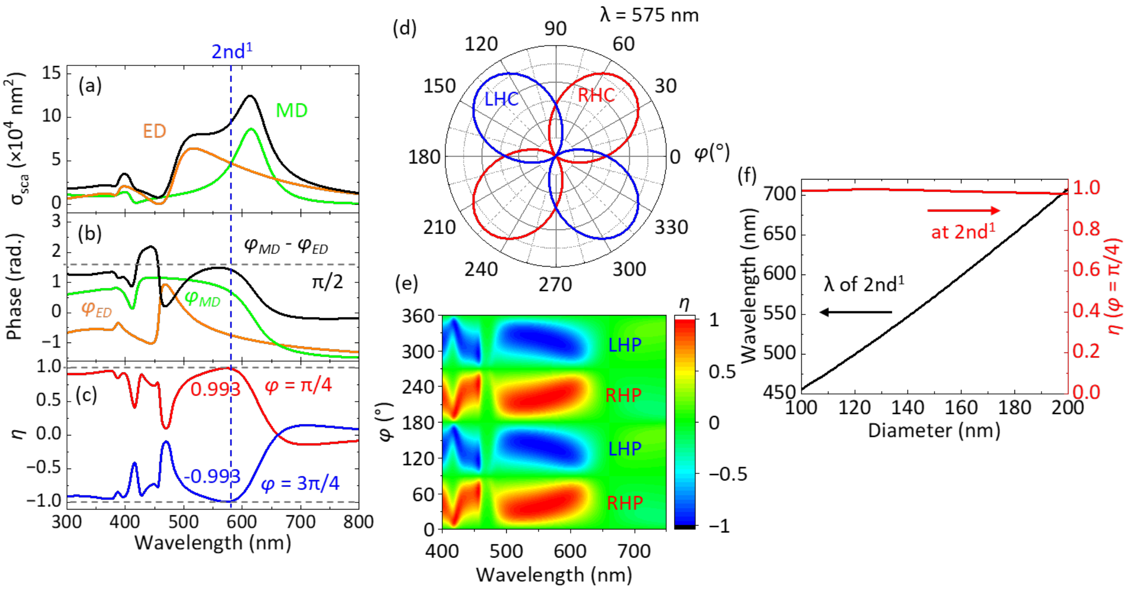

Figure 3. a) Scattering cross section, b) phase and c) degree of circular polarization $\eta(\theta = \pi/2, \varphi = \pi/4, 3\pi/4)$ spectra of a silicon nanosphere 150 nm in diameter in water (n_m=1.33). d) Scattering radiation patterns of RHC and LHC polarized light at the wavelength of 2nd^1 (575 nm). e) The degree of circular polarization as a function of the wavelength and angle φ ($\theta = \pi/2$). f) Wavelength of 2nd^1 (black) and $\eta(\theta = \pi/2, \varphi = \pi/4)$ (red) as a function of the diameter of a silicon nanosphere.

3. Experimental method

We now move on to experimental demonstration of circularly polarized scattering radiation from silicon nanospheres. Since a substrate that supports a silicon nanosphere disturbs the polarization state of scattered light, we employ silicon nanospheres suspended in water as a test sample. In this case, we can measure spectra of light scattered in any directions. A possible problem is the multiple scattering, but this can be avoided by using very diluted solutions. Another possible problem is the effect of size distribution. In Supporting Information (Figure S3), we show the scattering cross section spectra, the phase spectra, and the spectra of the degree of circular polarization in the $\varphi = \pi/4$ and $3\pi/4$ directions ($\theta = \pi/2$) for silicon nanospheres with different size distributions. Although the spectra are broadened and $\eta(\theta = \pi/2, \varphi = \pi/4)$ decreases with increases the size distribution, $\eta > 0.95$ is possible if the coefficient of size variation (c_v) defined as the standard deviation (σ) divided by the average diameter (D_{ave}) is smaller than ~5%.

Silicon nanospheres used for the experiments were prepared by thermal disproportionation of silicon monoxide (SiO) powder and subsequent hydrofluoric acid etching[31,32]. The detailed procedure is shown in Supporting Information ("Preparation of water solution of colloidal silicon nanospheres."), and results of the detailed structural characterizations are found in our previous papers [30,31]. **Figure 4**(a) shows the photos of water solutions of different size silicon nanospheres. Below the photos, D_{ave} and σ obtained from the extinction spectra are shown[31]. The average diameter is changed from 111.3 nm to 165.9 nm, and the c_v is ~5% in all the samples. Figure 4(b) and (c) shows typical TEM images of silicon nanospheres. The nanospheres are almost perfectly spherical and have high crystallinity, and the size distribution is very narrow.

Figure 4(d) shows the experimental setup for the measurement of φ angle-resolved circular polarized components in a scattering spectrum. The concentration of silicon nanospheres in a quartz cuvette (1 cm×1 cm) is set to be very low (~5×10^8 /mL) to avoid multiple scattering. The cuvette is illuminated by collimated linearly polarized light from a tungsten-halogen lamp (HORIBA, LSH-T250). Light scattered in $\theta = 90° \pm 4.7°$ direction is sent to a spectrometer via a quarter wave plate and a linear polarizer to measure the intensities of the RHC and LHC-polarized components of scattered light. To measure the φ dependence of scattering spectra, the linear polarizer on the incident side is rotated. The relation between the polarizer angle and the angle φ of scattered light is shown in Figure 4(f). The background signal arising from residual multiple scattering and reflection from cuvette walls is removed by the procedure shown in Supporting Information (Figures S4 and S5).

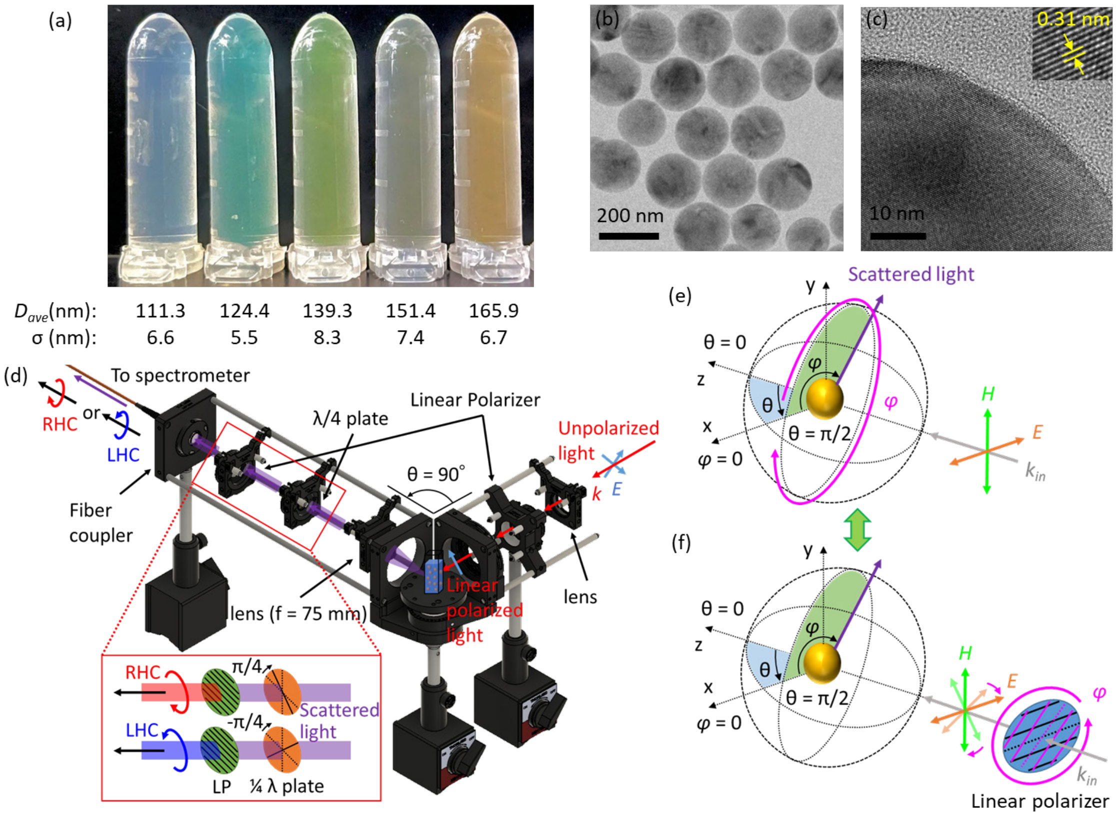

Figure 4. a) Photos of water solutions of silicon nanospheres with different diameters. The average diameter (D_{ave}) and the standard deviation (σ) are shown below the images. b) TEM image of silicon nanospheres and c) enlarged TEM image of a single silicon nanosphere. Inset shows the high-resolution TEM image. d) Setup for the circular polarization-resolved scattering measurements under linearly polarized light illumination. e) Definition of the scattering angle φ. f) Relation between the angle of the linear polarizer and the scattering angle φ defined in e). The angle φ of the linear polarizer (magenta) that determines the polarization state of the incident light is defined counterclockwise from the x-axis. The scattering angle φ defined in e) and the rotation angle φ of the polarizer are equivalent.

4. Results and discussion

Figure 5(a) shows the scattering spectra of the ED and MD modes in the $\varphi = \pi/4$ direction separately measured by the method shown in Ref. [33], and Figure 5(d) shows the calculated spectra. The average diameter of silicon nanosphere is 151.4 nm and the standard deviation of the size distribution is 7.4 nm. The size distribution is taken into account in the calculation. The good agreement between the measured and calculated spectra confirms the high quality of the silicon nanosphere sample. Figure 5(b) shows RHC (I_{RHC}) and LHC (I_{LHC}) circularly polarized components of scattered light in the φ = π/4 direction measured by the setup in Figure 4(d). Significantly different spectra are obtained between the RHC and LHC spectra. The spectra again agree very well with the calculated ones (Figure 5(e)). Figure 5(c) shows the degree of circular polarization obtained from Figure 5(b). The degree of circular polarization has a broad peak around 570 nm. The maximum value is close to 0.8. Therefore, silicon nanospheres in water radiate circularly polarized light in the φ = π/4 direction under linearly polarized plane wave illumination.

Figure 5(g) and (h) shows the measured and calculated, respectively, scattering radiation patterns of RHC and LHC polarized components at 570 nm. As expected, RHC polarized light is dominant at φ = π/4, 5π/4, and LHC polarized light is dominant at φ = 3π/4, 7π/4. Figure 5(i) and (j) shows the measured and calculated, respectively, degree of circular polarization as functions of φ and wavelength. Circularly polarized light is radiated in the 500-600 nm wavelength range.

In the calculated spectra in Figure 5(j), circularly polarized light is also emitted below 450 nm. However, it is not observed in the measured spectra in Figure 5(i). The discrepancy is due to the fact that only the lowest-order MD and ED modes are considered in the calculations, and that contributions of the higher-order modes in actual samples degrade η in the short wavelength range.

It should be stressed here that the observed radiation of circularly polarized light under linearly polarized plane wave illumination is not possible in spherical plasmonic nanoparticles that have only the ED mode. The date equivalent to Figure 5 obtained for gold nanospheres are shown in Supporting Information (Figure S6). In the case of gold nanospheres, the spectra and radiation patterns of RHC and LHC components are overlapped. Therefore, the scattered light is always linearly polarized.

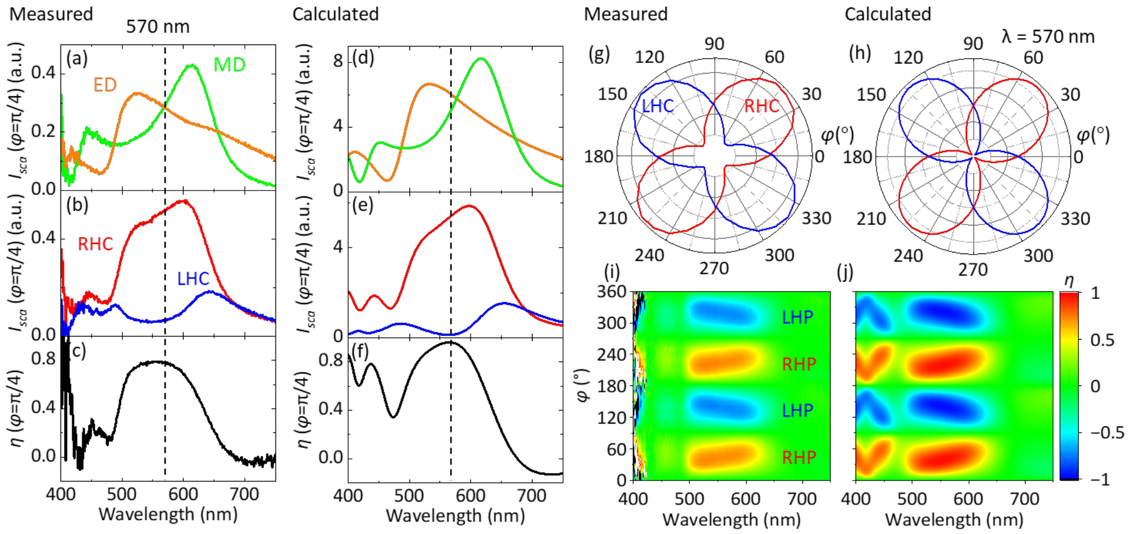

Figure 5. a-c) Measured and d-f) calculated scattering spectra of silicon nanosphere solution (D_{ave} = 151.4 nm, σ = 7.4 nm) in $\varphi = \pi/4$ direction. a,d) The ED and MD components and b,e) RHC and LHC polarized components of the scattering spectrum. c,f) the degree of circular polarization. g) Measured and h) calculated scattering radiation patterns of the RHC and LHC polarized components at 2nd[1] (570 nm). i) Measured and j) calculated $\eta(\theta = \pi/2)$ as functions of φ and wavelength.

In Figure 6, the spectra of $\eta(\theta = \pi/2, \varphi = \pi/4, 3\pi/4)$ obtained for different size silicon nanospheres (D_{ave} = 111.3-165.9 nm) are shown together with the calculated ones. The size distribution is taken into account in the calculations. The corresponding RHC and LHC polarized components of scattered light are shown in Supporting Information (Figures S7 and S8). The wavelength range of circularly polarized light emission shifts to longer wavelength with increasing the diameter because of the shift the 2nd Kerker condition.

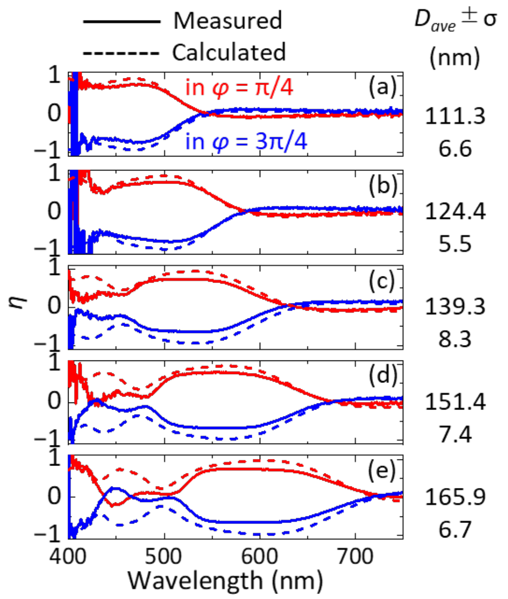

Figure 6. (solid line) Measured and (dashed line) calculated $\eta(\theta = \pi/2, \varphi = \pi/4, 3\pi/4)$ spectra of solutions containing different size silicon nanospheres. The average diameter (D_{ave}) and the standard deviation (σ) are shown on the right.

5. Summary

We have shown theoretically and experimentally that a silicon nanosphere can radiate circularly polarized light in specific directions under linearly polarized plane-wave illumination in water due to the interference between the orthogonal ED and MD modes. The idea to make it possible is controlling the phase difference between the ED and MD modes by the refractive index of a surrounding medium. It was shown that if the refractive index of a surrounding medium (n_m) is around 1.3, the phase difference becomes $\pi/2$, and circularly polarized light is radiated in specific directions. The theoretical prediction was experimentally confirmed for water solutions of size-purified silicon nanospheres. The demonstration of circularly polarized radiation in spherical nanoparticles under linearly polarized plane wave illumination suggests that an achiral nanosphere can be a nanoantenna for tiny circularly polarized light source.

Supporting Information
Supporting Information is available from the Wiley Online Library or from the author.

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